

Ionic-Liquid-Based Aqueous Two-Phase Interfacial Polymerization of Covalent Organic Framework Membranes for Molecular Separation

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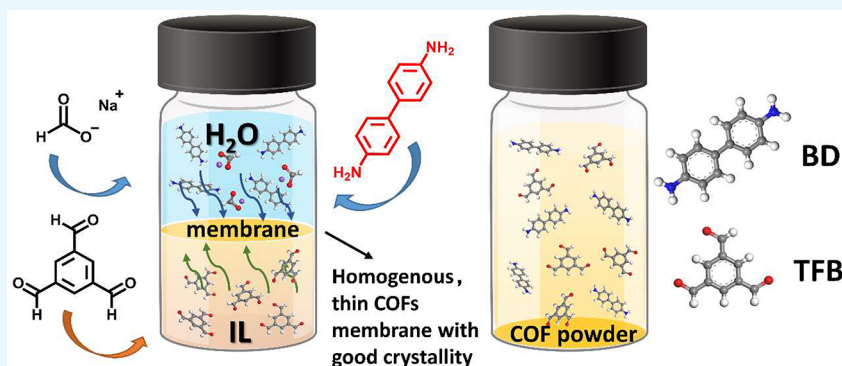
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ABSTRACT: Ionic liquid (IL)-based aqueous two-phase systems (ATPSs) are formed by the phase separation of aqueous mixture of ILs. Owing to their unique interfacial properties and stable interfacial zone, IL-based ATPSs can work as a promising confined space for membrane processing. Herein, for the first time, we proposed the thermoresponsive IL-based ATPSs as a platform for synthesizing covalent organic framework (COF) membranes. Upon the formation of aqueous biphasic interface at a corresponding temperature, the monomers and catalysts were separately dispersed in the IL-rich phase and water-rich phase. Due to the high viscosity of IL-rich phase and hydrogen bonding interactions in water-rich phase, the diffusion of both aldehyde and amine monomers could be controlled simultaneously and react within the aqueous biphasic interface zone. By optimizing the interfacial properties, the tight and crystalline COF membranes can be achieved with ultrahigh solvent flux and exceptional dye retention performance (>98.0%). For comparison, the homogeneous IL solution realized by temperature without affecting the integrity of the biphasic system was also applied as the medium for COF synthesis and only COF powders with low crystallinity can be obtained. As the pioneering exploitation to fabricate COF membrane at the IL-based aqueous biphasic interface, this work would supply a guiding direction for the green membrane processing strategy.

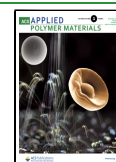
KEYWORDS: Ionic liquid, Aqueous two-phase system, Covalent organic framework membrane, Interfacial polymerization, Molecular separation

1. INTRODUCTION

Covalent organic framework (COF) membranes, which display a remarkably crystalline structure, high porosity, uniform pore size, and good stability, can be recognized as a considerable candidate for membrane separation application.^{1–4} Currently, several methods have been reported for preparing COF membranes, including both top-down (such as mechanical force- and surface-assisted assembly of COF precursors)^{5,6} and bottom-up (including polymerization at the interfaces of gas–liquid, gas–solid, and liquid–liquid) strategies.^{7–13} Among these techniques, the interfacial polymerization, especially liquid–liquid interfacial polymerization (usually organic solvent–water interface), with its unique designability, scalability, and controllability, has become a favored approach to synthesize COF membranes.^{14–16} The crucial factor to fabricate homogeneous and crystalline COF membranes is the

control of the polymerization reaction and/or the self-assembly of COF nanoparticles, which can be realized by reducing the diffusion rate of monomers or catalysts when dissolved in different liquid phases.^{17–20} However, in order to construct a liquid–liquid interface, the use of organic solvents is always inevitable. The evaporation of volatile organic solvents during the reaction always leads to an unstable interface. Additionally, the diffusion of monomers or catalysts would be uncontrolled in organic solvents, and the region size as well as the interfacial

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tension of the organic solvent–water interface are difficult to regulate. These issues often result in a loose stacking membrane structure formed at the organic solvent–water interface, affecting the crystallinity of the COF membranes. Thus, developing an all-aqueous interface with tunable interfacial properties can not only meet the environmental requirements but also regulate the growth process of the COF membranes.

Aqueous two-phase systems (ATPSs), commonly referred to as all-aqueous systems, are arisen from the phase separation of aqueous solution containing two incompatible solutes, such as two incompatible polymers or one polymer and other salts.²¹ The two aqueous phases are predominantly composed of water and can be endowed with particular biocompatibility owing to the independence from organic solvents. Furthermore, the permeability of the aqueous–aqueous interface (or denoted as water–water interface) can allow the free diffusion and exchange of small molecules and water, which in turn affect the interfacial properties.²² Compared with common organic solvent–water interface (or denoted as oil–water interface), the interfacial tension of water–water interface can be tunable and as low as 10^{-4} – 10^{-1} mN·m⁻¹, which is several orders of magnitude lower than that of oil–water interface.²³ These unique properties of ATPSs would create new opportunities in the fabrication of COF membranes. Very recently, Jiang and co-workers reported a water–water interfacial polymerization strategy to synthesize COF membranes via an ATPS formed by two incompatible polymers, poly(ethylene glycol) (PEG) and dextran (Dex).²⁴ With the distribution of amino and aldehyde monomers into the two aqueous phases, a highly crystalline COF membrane can be achieved under the optimized interfacial conditions, exhibiting high NaCl rejection and water permeance for desalination. It should be noted that the bulk polymers (PEG or Dex) which could regulate the assembly of COF units due to the steric hindrance may persist within the COF membranes and require rigorous washing to eliminate the residuals.²⁴ Therefore, it is essential to simplify the postprocessing of COF membranes by using an alternative ATPS composed of small molecules, which can be easily removed.

Ionic liquids (ILs) are a kind of organic salt comprising solely of ions with the melting point below 100 °C.²⁵ By virtue of their distinctive ionic structures, ILs exhibit remarkable properties such as low volatility, negligible flammability, high thermal and chemical stability, strong dissolving ability, and feasible structural designability.^{26–28} IL-based ATPSs are biphasic systems created by the hydrophobic ILs or hydrophilic ILs with a certain concentration of inorganic salts.^{29–31} The biphasic state is composed of an IL-rich phase and a water-rich phase, divided by a relatively sharp interface. Especially for some hydrophobic ILs, the phase transition between a homogeneous solution to aqueous biphasic can be driven by temperature. The phase separation occurs either with decreasing temperatures, known as an upper critical solution temperature (UCST), or less often, with increasing temperatures, known as a lower critical solution temperature (LCST).^{32–34} With the reversible switching between construction and deconstruction of the interface, IL-based ATPSs, which combine the environmental friendliness of ILs and the structural superiority of ATPSs, may offer a promising interfacial platform for the fabrication of COF membranes.

Herein, we developed thermoresponsive IL-based ATPSs as an interfacial strategy to fabricate COF membranes from a

Schiff base reaction for the first time. The IL-based ATPSs were formed by three different ILs with the same cations, tetra-*n*-butylphosphonium 2,4-dimethylbenzenesulfonate ([P₄₄₄₄]-[DMBS]), tetra-*n*-butylphosphonium maleate ([P₄₄₄₄][Ma]), and tetra-*n*-butylphosphonium fumarate ([P₄₄₄₄][Fum]), in which the former two ILs were LCST-type and the last one was UCST-type. Upon the formation of the aqueous biphasic interface at a corresponding temperature, the distributed monomers and catalysts in the two aqueous phases would diffuse and react within the interface zone. By optimizing the interfacial properties, tight and intact COF membranes can be achieved. For comparison, the homogeneous state realized by temperature without affecting the integrity of the biphasic system was also applied as the medium for the COF synthesis.

2. EXPERIMENTAL SECTION

2.1. Materials and Chemicals. Benzene-1,3,5-tricarboxaldehyde (TFB, 97%, Aladdin), 2,4,6-Triformylphloroglucinol (Tp, 97%, Aladdin), benzidine (BD, 95%, Aladdin), sodium formate (99.5%, J&K Scientific, Ltd.), sodium acetate (99.5%, J&K Scientific, Ltd.), and sodium phosphate dibasic (99%, J&K Scientific, Ltd.) were employed directly in the synthesis of COF without further purification. Tetra-*n*-butyl phosphonium hydroxide ([P₄₄₄₄][OH], 40 wt % in water, Aladdin), *m*-xylene-4-sulfonic acid hydrate (DMBS, 98%, TCI), maleic acid (Ma, 99%, J&K Scientific, Ltd.), and fumaric acid (Fum, 99%, J&K Scientific, Ltd.) were directly used in the synthesis of ILs with no refinement. Poly tetra fluoroethylene (PTFE) microfiltration membranes, with an average pore size of 0.22 μm, were purchased from Tianjin Branch billion Lung Experimental Equipment Co., Ltd. Alcian blue (AB, 98%, Macklin), brilliant blue R (BBR, 98%, Macklin), methyl blue (MB, 98%, Macklin), and rhodamine B (RhB, 98%, Macklin) were used as dyes for separation. Deionized water was employed in all experimental procedures.

2.2. Characterization. ¹H NMR spectra of the ILs were obtained by using a Bruker Advance 400 NMR spectrometer. Powder X-ray diffraction (PXRD) patterns were obtained in the range of $2\theta = 2$ – 40° with a scan speed of $1^\circ\cdot\text{min}^{-1}$ on a Rigaku SmartLab 9 kW X-ray diffractometer (Cu K α radiation). Fourier transform infrared (FT-IR) spectra were analyzed by a Bruker Tensor II spectrometer in the wavenumber range 4000–400 cm⁻¹. Nitrogen adsorption and desorption isotherms were recorded at 77 K on a multistation extended specific surface area and porosity analyzer (ASAP 2460–4MP). Before gas adsorption treatment, the samples were degassed under vacuum at 150 °C for 12 h. Surface areas were calculated by the Brunauer–Emmett–Teller (BET) model applied between P/P_0 values of 0.05 and 0.3 for microporous and mesoporous COFs. Corresponding pore size distributions were calculated using the nonlocalized density functional theory (NLDFT), and the presence of wide range of pores was determined using Barrett–Joyner–Halenda (BJH) analysis. The morphologies of COFs were obtained using a scanning electron microscope (Gemini SEM 300) operating at 10 kV with tungsten filament as the electron source. Transmission electron microscopy (TEM) images were captured on JEM-2100F (JEOL) TEM at an accelerating voltage of 200 kV. The proton concentration of water-rich phase was measured by a Mettler Toledo FE28-Standard pH meter. The density of ILs was determined at 25 °C under atmospheric pressure by MDJ-300S electronic densimeter with a precision of ± 0.0001 g·cm⁻³. The viscosity of ILs was measured under atmospheric pressure on a Haake Rheostress 6000 rheometer with a vertebral plate system (C35/ITiL) and a Circulator HAAKE DC10 circulating water system at a controlled temperature of 25 or 41 °C. Interface tension measurements were performed on a Sigma 700 interfacial tensiometer by the Du Noüy method with a precision of ± 0.001 mN·m⁻¹. Zeta potential of TFB-BD-DMBS(I) and Tp-BD-DMBS(I) membrane was measured at pH 7 using a Zeta potential analyzer for solid surface (SurPASS Anton Paar).

2.3. Synthesis of ILs. The ILs employed in this study, namely, tetra-*n*-butylphosphonium 2,4-dimethylbenzenesulfonate ([P₄₄₄₄]-

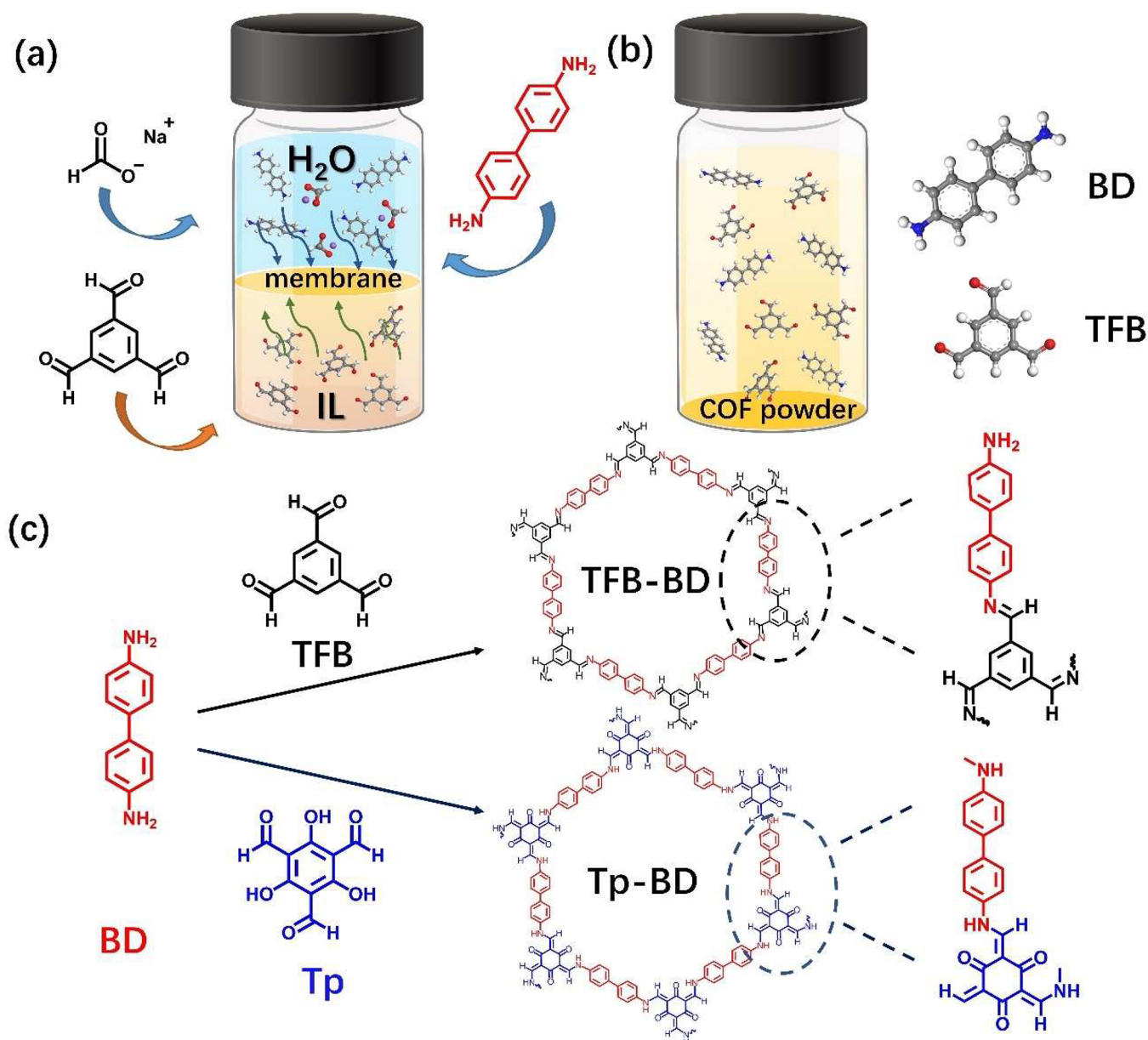


Figure 1. Schematic diagram of the synthesis of TFB-BD and Tp-BD COF using two different methodologies. (a) Thin-film formation at the IL-based aqueous biphasic interface under the phase-separation temperature. The lower-phase denoted the aldehyde monomers dissolved in IL-rich phase, and the upper-phase corresponded to the amine monomers in water-rich phase. (b) Powder formation in the homogeneous IL aqueous solution at the monophasic temperature. (c) Chemical structures of TFB-BD and Tp-BD COFs.

[DMBS]), tetra-*n*-butylphosphonium maleate ([P₄₄₄₄][Ma]), and tetra-*n*-butylphosphonium fumarate ([P₄₄₄₄][Fum]), were synthesized and purified following the methodologies in previous works.^{35,36} To prepare the ILs, 0.0715 mol of acid (13.32 g of DMBS, 8.30 g of Ma and 8.30 g of Fum) was dissolved in ultrapure water, and an equivalent amount of 40 wt % [P₄₄₄₄][OH] aqueous solution (50 mL) was added. The mixture was then stirred at room temperature for 24 h. The postprocessing for the three ILs was different. For [P₄₄₄₄][DMBS] and [P₄₄₄₄][Ma], the resulting product was extracted with dichloromethane and subjected to three water washes. For [P₄₄₄₄][Fum], the product underwent evaporation to remove the water and recrystallized three times in an ethyl acetate–ethanol mixture. Finally, the crude product was vacuum-dried at 60 °C for 48 h. The yields for the ILs synthesized were 28.69 g for [P₄₄₄₄][DMBS] (86%), 26.03 g for [P₄₄₄₄][Ma] (92%), and 25.50 g for [P₄₄₄₄][Fum] (90%). The ¹H NMR spectroscopy of the ILs were shown in Figure S1.

[P₄₄₄₄][DMBS]: δ = 7.80 (d, 1H), 6.91 (s, 1H), 6.64 (d, 1H), 2.24 (s, 3H), 2.27–2.14 (m, 8H), 2.24 (s, 3H), 1.43–1.36 (m, 16H), 0.88 (t, 12H). [P₄₄₄₄][Ma]: 0.86 (t, 12H), 1.38–1.25 (m, 16H), 2.15 (m, 8H), 6.09 (s, 2H). [P₄₄₄₄][Fum]: 0.91 (t, 12H), 1.37–1.50 (m, 16H), 2.20 (m, 8H), 6.38 (s, 2H).

2.4. Fabrication of IL-Based ATPSs. Based on our previous works,^{37,38} the IL-based ATPSs were formulated by the mixture of ILs and water. The critical dissolution temperature was determined by continuous heating within a temperature-controlled water bath. First, a certain amount of IL and water were mixed and stirred for 2 h. Then, these mixtures were placed in an ice water bath to reach 0 °C and gradually heated to 60 °C. The lower critical dissolution temperature (LCST) was determined as the cloud point when heating. The aqueous solutions of [P₄₄₄₄][DMBS] and [P₄₄₄₄][Ma] exhibited a remarkable transition from transparent to opaque with the increasing temperature, demonstrating the LCST-type phase transition. In contrast, the aqueous solution of [P₄₄₄₄][Fum] changed

from opaque to transparent when the temperature was increased, which involved the UCST-type phase transition. The UCST was the temperature when the mixture changes from opaque to transparent during heating.

2.5. Synthesis of COF Membrane at IL-Based Aqueous Biphasic Interface. The COF membranes were formed by the reactions between amine monomer benzidine (BD) and two different aldehyde monomers, 1,3,5-benzenetricarboxaldehyde (TFB) or 2,4,6-triformylphloroglucinol (Tp). The mass fraction of ILs that we used was 26.3% for $[P_{4444}][DMBS]$, 21.9% for $[P_{4444}][Ma]$, and 26.0% for $[P_{4444}][Fum]$. Initially, a 4 mM aldehyde solution was obtained by dissolving 0.120 mmol of aldehyde (19.5 mg of TFB or 25.2 mg of Tp) in 30.0 mL of the IL. Subsequently, 0.120 mmol of BD (22.1 mg) and 0.120 mmol of sodium formate (1 equiv) were dissolved in 30.0 mL of deionized water to yield a 4 mM amine aqueous solution. For the single-activation of COF, no sodium formate catalyst was added in the amine solution. Then 2.0 mL of the aldehyde solution was added to a culture bottle as the lower-phase, followed by the addition of spacer water (5.4 mL of $[P_{4444}][DMBS]$, 7.0 mL of $[P_{4444}][Ma]$, and 5.1 mL of $[P_{4444}][Fum]$) on the top of aldehyde solution. Finally, 2.6 mL of BD aqueous solution was dropped on the top of the spacing solution as the upper-phase. The three systems were kept at the phase-separation temperature (25 °C for $[P_{4444}][Fum]$, 41 °C for $[P_{4444}][Ma]$, and $[P_{4444}][DMBS]$) for 144 h without disturbing the synthesis of COF membranes at the biphasic interface. The obtained membranes were collected and purified with water and ethanol for further experiments.

2.6. Synthesis of COF in the Homogeneous IL Solution. First, the aqueous solution of IL was obtained by blending 2.0 mL of IL with different amounts of deionized water (8.0 mL of $[P_{4444}][DMBS]$, 7.7 mL of $[P_{4444}][Ma]$, and 9.6 mL of $[P_{4444}][Fum]$). After that, 0.008 mmol of aldehyde (1.3 mg of TFB or 1.7 mg of Tp) and 0.010 mmol of BD (1.9 mg) were dissolved in the IL aqueous solutions. The solution was mixed thoroughly by ultrasound and vortex stirrer. The above three systems were kept at the single phase temperature (41 °C for $[P_{4444}][Fum]$, 25 °C for $[P_{4444}][Ma]$, and $[P_{4444}][DMBS]$) for 144 h. Finally, the COF products formed at the bottom of the solution were carefully collected through filtration and purified with water and ethanol prior to subsequent experiments.

2.7. Permeation and Rejection Evaluation. The permeation and rejection performance of the obtained COF membranes were evaluated by transferring the membranes to a PTFE microfiltration substrate. The solvent permeation of the COF membranes was investigated under an upstream pressure of 0.55 bar at 25 °C. Solvent flux values were determined by averaging triplicate experiments performed with three specimens. The value of permeance (P , $L \cdot m^{-2} \cdot h^{-1} \cdot bar^{-1}$) was calculated by the following equation:

$$P = \frac{V}{A \times t \times \Delta P} \quad (1)$$

where V was the solution volume permeating across the membrane (L), A was the effective membrane area (m^2) of the filtration cell, t was the operation time (h), and ΔP was the trans-membrane pressure (bar).

For the dye rejection measurements, aqueous solutions of dyes with a concentration of 50 μM were passed through the membrane coupons at a pressure of 0.55 bar. After discarding the first 1.0 mL of the filtrate, the following 5.0 mL of filtrate was gathered for the following analysis. The dye concentrations of the feed and permeate solutions were measured by UV-vis spectroscopy. The rejection rate (R , %) was calculated based on the following equation, in which C_F and C_P represented the dye concentrations of feed and permeate solutions.

$$R = \left(1 - \frac{C_P}{C_F} \right) \times 100\% \quad (2)$$

3. RESULT AND DISCUSSION

3.1. Interfacial Polymerization of COF Membranes.

First, the IL-based ATPSs were determined by the binary phase diagrams of ILs and water (Figure S2). The curves in phase diagram represented the phasic boundary between monophasic and biphasic. The mixtures of LCST-type ILs ($[P_{4444}][DMBS]$ and $[P_{4444}][Ma]$) and water transformed from a transparent solution to a biphasic system as the temperature increased. While for the UCST-type IL ($[P_{4444}][Fum]$), the separated phases turned into a single phase upon heating. As schematically exhibited in Figure 1a, the IL-based aqueous biphasic interface served as the platform for synthesizing COF membranes based on the Schiff base reaction involving amine monomer benzidine (BD) and aldehyde monomer 1,3,5-benzenetricarboxaldehyde (TFB) or 2,4,6-triformylphloroglucinol (Tp) (Figure 1c). The concentration of the IL and the corresponding critical solution temperature that we used were visually represented by the red triangle marker in the phase diagrams of three different IL-based ATPSs (Figure S2). In order to investigate the optimal reaction conditions for the synthesis of COF membrane in IL-based ATPSs, we used the $[P_{4444}][DMBS]$ -based ATPSs as a platform to synthesize TFB-BD COF membranes. First, three different reaction temperatures (41, 61, and 81 °C) which were all above the LCST of $[P_{4444}][DMBS]$ were selected for the interfacial polymerization. As shown in Figure S3a, only small COF fragments formed by loosely packed nanoparticles can be obtained at 61 °C, while for the reaction temperature of 81 °C, a negligible quantity of product was generated at the interface of $[P_{4444}][DMBS]$ -based ATPSs. Therefore, 41 °C was selected as the optimum temperature for the following interfacial polymerization. Second, the effect of reaction time on the interfacial synthesis was also explored. As shown in Figure S3b, the morphology of the TFB-BD membrane became flatter and denser as the reaction time increased, and a continuous and structured membrane can be obtained when the reaction time reached 144 h. Based on the above preliminary results, the typical interfacial synthesis protocols were as follows. A 4 mM aldehyde solution was first obtained by dissolving TFB in $[P_{4444}][DMBS]$ as the bottom phase. After the addition of a water spacer layer, a 4 mM aldehyde aqueous solution containing 1 equiv of sodium formate was carefully introduced to obtain the top phase. Then the biphasic system was kept at the corresponding phase-separation temperature (41 °C for $[P_{4444}][DMBS]$) under an undisturbed condition for 144 h. During the reaction period, the aldehyde monomer in the IL-rich phase was activated by $[P_{4444}][DMBS]$ (a Lewis acid), and the amine monomer in the water-rich phase was activated by sodium formate (a Lewis base). After dual-activation, the amine and aldehyde monomers diffused oppositely and reacted at the biphasic interface to fabricate a yellow, continuous, and free-standing membrane defined as TFB-BD-DMBS(I). In contrast, the TFB-BD COF was also synthesized in homogeneous phase at the corresponding temperature (25 °C for $[P_{4444}][DMBS]$) under the same condition (4 mM monomers, 144 h), and only powder product was obtained and denominated as TFB-BD-DMBS(H) (Figure 1b). To identify the general applicability and versatility of the IL-based aqueous two-phase interfacial polymerization of COF membranes, the biphasic interfaces of $[P_{4444}][Ma]$ -H₂O and $[P_{4444}][Fum]$ -H₂O systems were also constructed, i.e., the solvent $[P_{4444}][DMBS]$ was exchanged for the other ILs ($[P_{4444}][Ma]$ and $[P_{4444}][Fum]$), and achieved the membrane

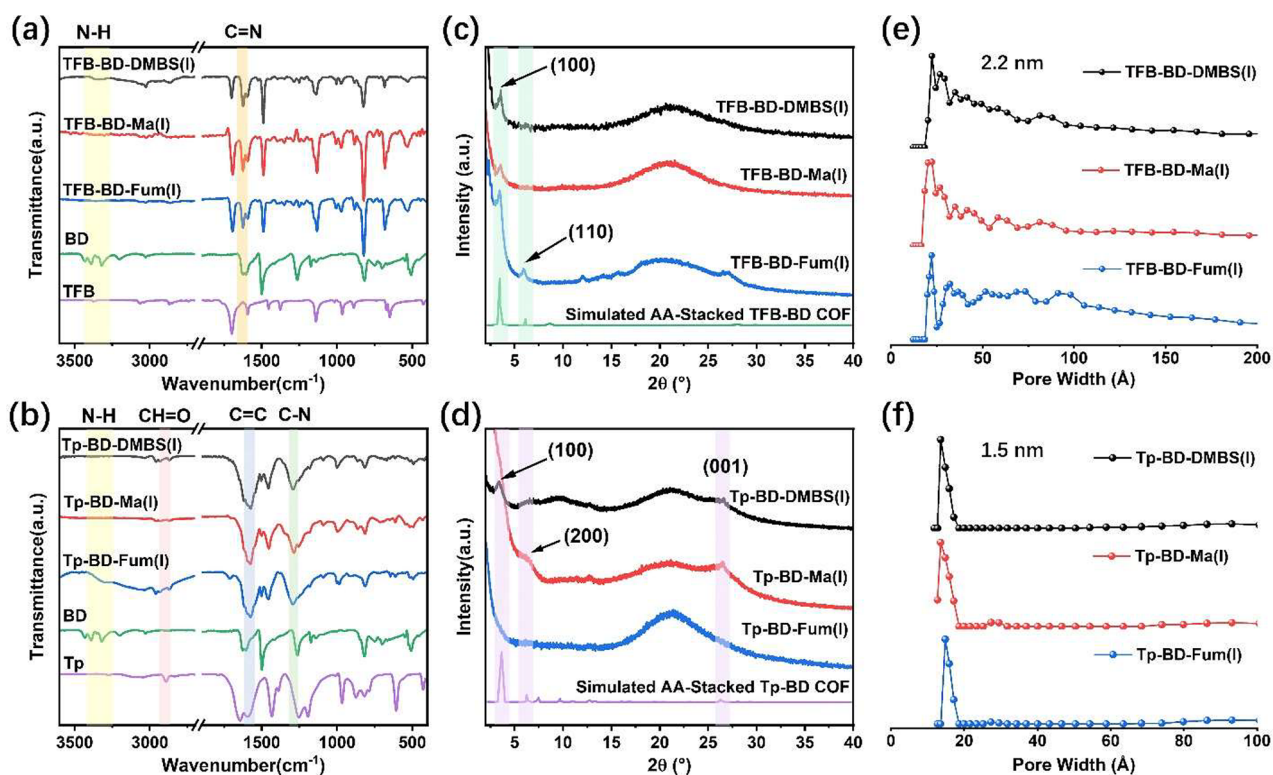


Figure 2. Structural characterization of the COF membranes. Comparison of the FT-IR spectra of (a) TFB-BD COF and (b) Tp-BD COF synthesized at different IL-based aqueous biphasic interface. Comparison of the PXRD patterns of (c) TFB-BD COF and (d) Tp-BD COF synthesized at different IL-based aqueous biphasic interface. Pore size distribution of (e) TFB-BD COF and (f) Tp-BD COF synthesized at different IL-based aqueous biphasic interface.

products. The specified procedures for the synthesis of COF are provided in the [Experimental Section](#), and the corresponding product state and denotation are listed in [Table S1](#) and [Figures S4](#) and [S5](#).

3.2. Characterization of COF Membrane. Fourier transform infrared (FT-IR) spectra were used to examine the chemical composition of the obtained COF membranes. As shown in [Figure 2a](#), new stretching bands at 1630 cm^{-1} confirmed the successful establishment of the imine linkage of TFB-BD COF,³⁹ while for Tp-BD COF, new stretching bands at 1280 and 1550 cm^{-1} , corresponding to C–N and C=C, confirmed the presence of β -ketoenamine linkage ([Figure 2b](#)).⁴⁰ Furthermore, the absence of characteristic peaks of $-\text{NH}_2$ and $-\text{CHO}$ at 3100 and 2900 cm^{-1} demonstrated complete conversion of the starting monomers. The crystalline structure of the as-synthesized COF membranes was examined using powder X-ray diffraction (PXRD). As shown in [Figure 2c](#), all the TFB-BD COF membranes formed at the IL-based aqueous biphasic interface displayed a sharp peak at a lower 2θ value of 3.5° , which was assigned to the reflection of the (100) plane and consistent with the simulated AA eclipsed stacking model. The COF membrane TFB-BD-Fum(I) which fabricated by dual-activation interfacial polymerization in $[\text{P}_{4444}]\text{-[Fum]-H}_2\text{O}$ system at 25°C exhibited another diffraction peak with 2θ of 6.0° , which could be attributed to the (110) plane.⁴¹ Similarly, all the Tp-BD COF membranes displayed minor peaks at 6.2° and 26.0° , attributing to the (200) and (001) reflection planes, respectively.⁴² Compared with other Tp-BD COF membranes ([Figure 2d](#)), only Tp-BD-DMBS(I) exhibited a sharp peak at 3.5° , indicating the highly crystalline and long-range ordered framework of the COF membrane,

while for all of the COF powders synthesized in homogeneous IL solution at the corresponding monophasic temperature, no obvious reflection peaks can be observed in PXRD spectra, indicating the poor crystalline structure ([Figure S7](#)). To elucidate the influence of monomer activation on the fabrication of COF membranes, the interfacial polymerization with single-activation (without sodium formate in amine aqueous solution) was also carried out, and only brittle films could be obtained. The weak (100) characteristic peaks for the single-activated COF films indicated their low crystallinity ([Figure S7](#)).

Nitrogen adsorption–desorption experiments were applied to determine the porosity of the COF membranes. As illustrated in [Figures S8](#) and [S9](#), both the TFB-BD and Tp-BD COF membranes exhibited a relatively low Brunauer–Emmett–Teller (BET) surface area compared to previous works, such as $99.50\text{ m}^2\text{g}^{-1}$ for TFB-BD-DMBS(I) and $102.73\text{ m}^2\text{g}^{-1}$ for Tp-BD-DMBS(I).^{43–46} We proposed that the irregular assembly of COF crystallite within the membranes may generate a highly staggered interlamellar pores, which restricted the adsorption of nitrogen and led to the low surface areas.²⁴ In addition, compared with the COF powders synthesized in monophasic, all the COF membranes showed a narrow pore-size distribution, which TFB-BD(I) was 2.2 nm and Tp-BD(I) was 1.5 nm ([Figure 2e,f](#)), further indicating the interlaced assembly of COF crystallite within membranes.

Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) were employed to ascertain the morphologies of the COF membranes. The three TFB-BD interfacial polymerized COF membranes presented similar surface morphologies. The TFB-BD-DMBS(I) was assembled

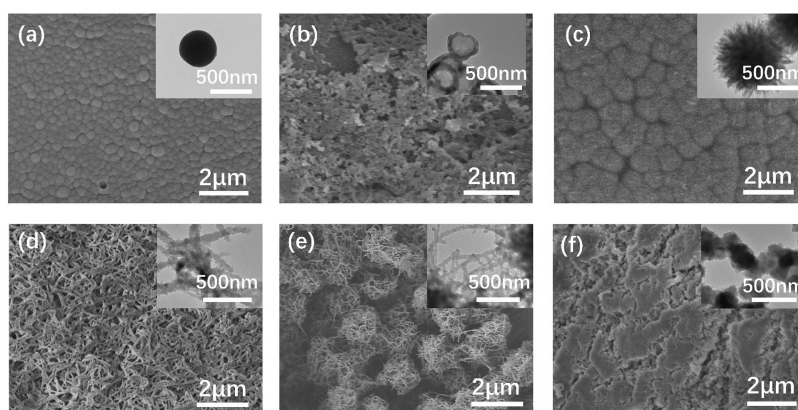


Figure 3. Top-view SEM images of (a) TFB-BD-DMBS(I), (b) TFB-BD-Ma(I), (c) TFB-BD-Fum(I), (d) Tp-BD-DMBS(I), (e) Tp-BD-Ma(I), and (f) Tp-BD-Fum(I); the insets are TEM images.

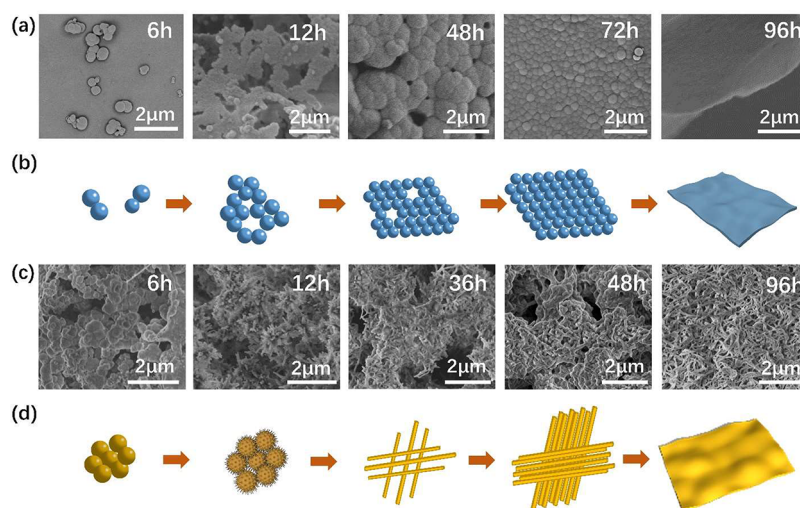


Figure 4. Schematic representation of the mechanism of membrane formation. (a) SEM images of the TFB-BD-DMBS(I) growth process. (b) Schematic diagram of TFB-BD-DMBS(I) film formation. (c) SEM images of the Tp-BD-DMBS(I) growth process. (d) Schematic diagram of Tp-BD-DMBS(I) film formation.

by dense arrangement of uniform and smooth COF particles with diameters smaller than 500 nm (Figure 3a). While for TFB-BD-Ma(I), it could be observed that only irregular COF particles loosely stacked, which resulted in a more brittle and cracked membrane (Figure 3b). The TFB-BD-Fum(I) was formed by the assembly of spiny COF particles, approximately 800 nm in diameter (Figure 3c). Moreover, a different morphology was observed for the Tp-BD COF membranes. The Tp-BD-DMBS(I) and Tp-BD-Ma(I) were fabricated by wormlike COF fibers (Figure 3d,e). Similarly, Tp-BD-Fum(I) was also composed of COF fibers, which were joined together to arrange wider tabular structures (Figure 3f), while for the COF powders formed at the homogeneous IL solutions (Figure S10), only random stacking of irregular COF particle with a diameter about 300 nm for TFB-BD and 100–200 nm for Tp-BD can be observed, leading to the loose and fragmented COF structures.

3.3. Exploration of Membrane Formation Mechanism. To explore the crystallization process at the IL-based biphasic interface, TFB-BD-DMBS(I) and Tp-BD-DMBS(I) were used as examples to monitor the membrane formation by time-dependent SEM. It could be found in Figure 4a that TFB-BD-DMBS(I) displayed a spherical crystallite within the first 6 h of polymerization. With the reaction proceeding, the

spherical particles started to assemble in the $[P_{4444}][DMBS]$ -H₂O interface zone. At about 12 h, the globules stacked densely to form COF nanosheets; the nanosheets further connected laterally at 48 h and finally spread as a two-dimensional thin film at 72 h. The formation process of Tp-BD-DMBS(I) was similar to that of TFB-BD-DMBS(I) except for the morphology of crystallites (Figure 4c). After the formation of spherical particles at the first 6 h, the globules gradually transformed into spiny spheres at 12 h. Then the spiny particles were further assembled into wormlike nanofibers at 36 h. As the assembly proceeded, the COF nanofibers were intertwined at the $[P_{4444}][DMBS]$ -H₂O interface zone, and a membrane was achieved at 96 h.

Generally, the interfacial polymerization of COF membranes is dominated by three key factors: the reactivity of monomers, the diffusion reaction of monomers, and the assembly of crystallite within interfacial zone.²⁴ The activated monomers dissolved in two phases first diffuse oppositely toward the interface and then react with each other to form crystallites. The quality of crystallites is synergistically controlled by the kinetic process of monomer diffusion and thermodynamic process of bonding reaction between monomers. Upon the crystallites formed, the properties of the interface zone are crucial for the further spread and assembly of crystallites and

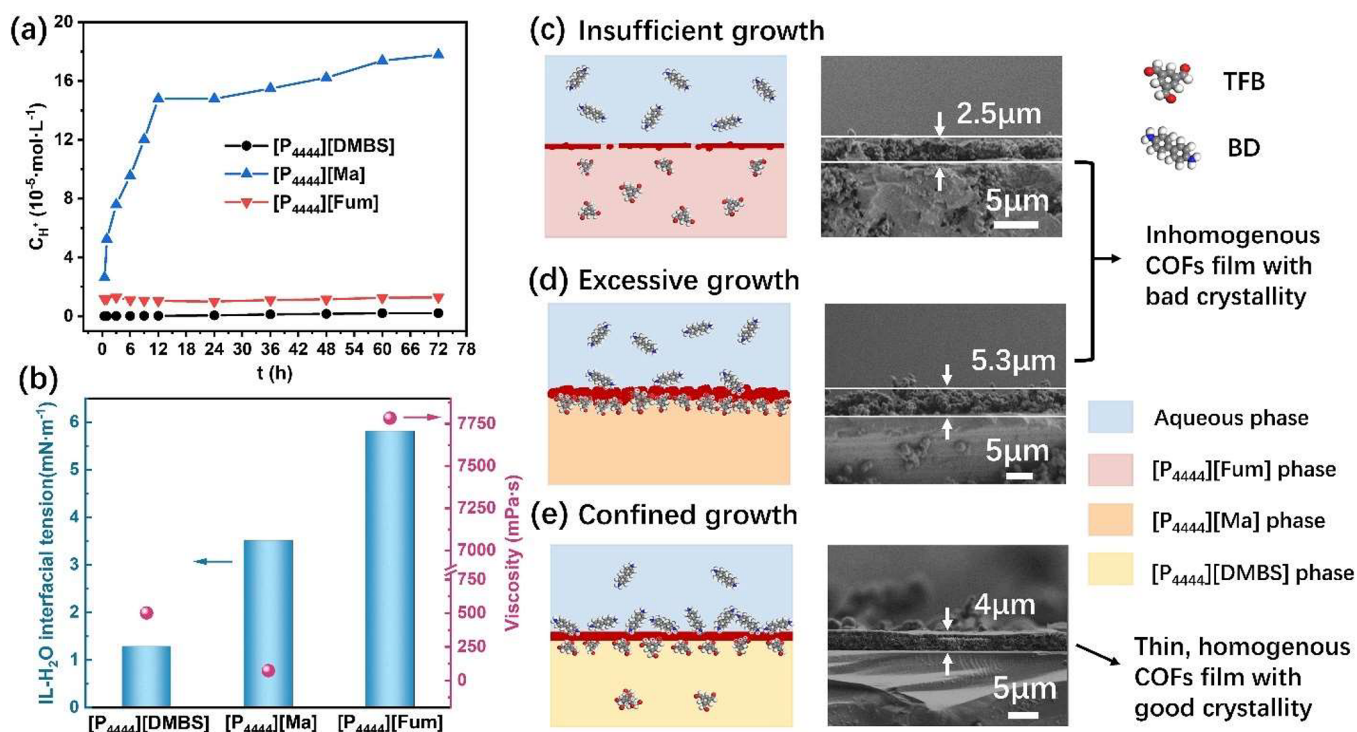


Figure 5. (a) Proton concentration of the water-rich phase for different ILs system as a function of reaction time. (b) Biphasic interfacial tension and viscosity of ILs at two phase temperature (41 °C for $[P_{4444}][\text{DMBS}]$ and $[P_{4444}][\text{Ma}]$, 25 °C for $[P_{4444}][\text{Fum}]$). Schematic for diffusion process of monomers in the bulk phase and the crystallization process at the IL-based aqueous biphasic interface (c) $[P_{4444}][\text{Fum}]$ -H₂O, (d) $[P_{4444}][\text{Ma}]$ -H₂O, and (e) $[P_{4444}][\text{DMBS}]$ -H₂O. The picture on the right is cross-sectional SEM images of the corresponding TFB-BD membrane.

determine whether to form membranes. Particularly, for the interfacial polymerization at IL-based biphasic interface, the pH values of the water-rich phase, the viscosity of ILs and the interfacial tension of biphasic interface were systematically investigated to explore their synergistic effects on the COF membrane formation. For Schiff-base reaction, the activity of aldehyde monomers can be enhanced by the acidic catalysts to convert $-\text{CHO}$ into $-\text{C}^+\text{HOH}$, which is more preferable to nucleophilic attack.⁴⁷ Thus, the intrinsic acidity of the three ILs $[P_{4444}][\text{DMBS}]$, $[P_{4444}][\text{Fum}]$, and $[P_{4444}][\text{Ma}]$ made them not only the bottom phase but also the acid activators of aldehyde monomers. While for the water-rich phase, the presence of small amount of ILs led to the increasing H^+ concentration, resulting in the protonation of amine monomers to transform $-\text{NH}_2$ into $-\text{NH}_3^+$ with limited nucleophilic attack ability.⁴⁸ Therefore, some activators to retard the protonation of amine monomers in the water-rich phase were required. As for the synthesis of TFB-BD-DMBS(I), three different amine monomer activators (sodium formate, sodium acetate, and sodium phosphate dibasic) with two different concentrations (1 equiv and 0.1M) were selected. The crystalline structures of the obtained COF membranes with different activators were verified by XRD. As shown in Figure S13, only the TFB-BD-DMBS(I) synthesized with 1 equiv of sodium formate displayed a sharp peak at a 2θ value of 3.5° , indicating the ordered COF structure with high crystallinity. By monitoring the time-dependent H^+ concentration of water-rich phase with 1 equiv of sodium formate as amine monomer activators (Figure 5a), it can be found that the acidity of water-rich phase for $[P_{4444}][\text{Ma}]$ -based ATPS was still at a high level, which would inhibit the reactivity of amine monomers. While for $[P_{4444}][\text{DMBS}]$ -based ATPS, the lowest H^+ concentration of the water-rich phase was beneficial

to achieve the activation of amine monomers. As a result, a continuous membrane can be obtained at the $[P_{4444}][\text{DMBS}]$ biphasic interface.

During the formation of crystallites, the physicochemical properties of ILs play a significant role in the diffusion and contact of monomers, which reduce the diffusion rate and promote the diffusion–reaction equilibrium to create a mild crystallite formation environment.⁴⁴ Except for the diffusion of BD monomers regulated by H-bonding interactions in the water-rich phase which was similar to the traditional interfacial polymerization at organic solvent–water interface, the diffusion of aldehyde monomers could also be controlled by the ILs with high viscosity simultaneously. As depicted in Figure 5b and Table S2, the values of viscosity for $[P_{4444}][\text{DMBS}]$, $[P_{4444}][\text{Fum}]$, and $[P_{4444}][\text{Ma}]$ at the selected synthesis temperature were 503.0, 7784.1, and 74.4 mPa·s, respectively. It was reasonable to propose that $[P_{4444}][\text{Ma}]$ with the lowest viscosity had the weakest limitation on aldehyde diffusion, resulting in a failed diffusion–reaction equilibrium within the interface zone and a final cracked film. The interface tension is another crucial factor to determine the polymerization process because that high interface tension is favorable to the anchorage of crystallites but impedes the further spread–assembly process, leading to the uneven and crumple structures.^{49,50} Thus, we examined the interface tension of the three IL-based aqueous biphasic at the corresponding phase-separation temperature and shown in Figure 5b and Table S2. The interface tension of $[P_{4444}][\text{DMBS}]$ -H₂O biphasic was $1.291 \text{ mN} \cdot \text{m}^{-1}$ at 41 °C, which was lower than that of $[P_{4444}][\text{Fum}]$ -H₂O biphasic ($5.817 \text{ mN} \cdot \text{m}^{-1}$) and $[P_{4444}][\text{Ma}]$ -H₂O biphasic ($3.516 \text{ mN} \cdot \text{m}^{-1}$). Considering the synergistic effect of ILs viscosity as well as the interface tension, $[P_{4444}][\text{Fum}]$ with the highest viscosity and interface

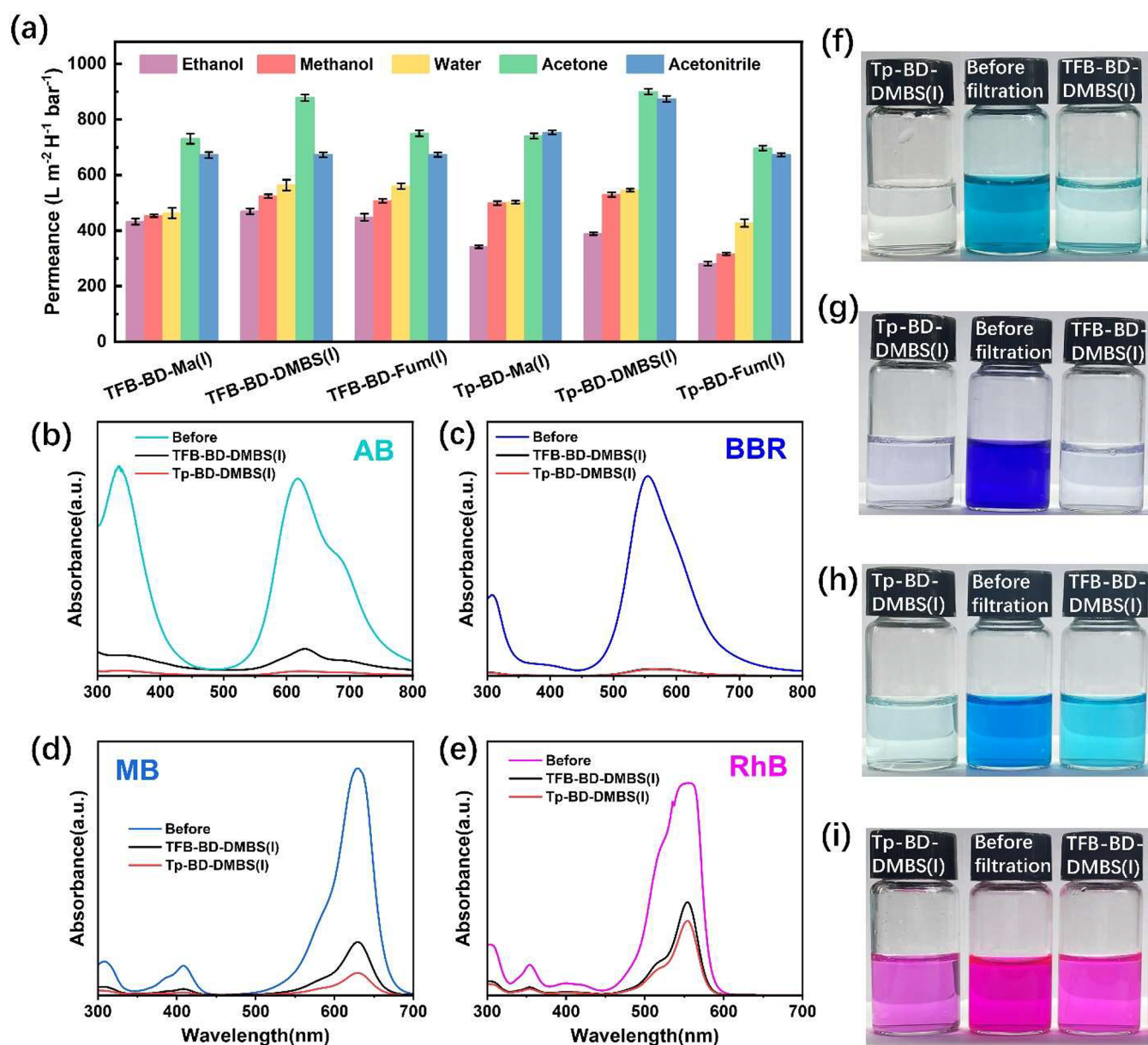


Figure 6. (a) Permeance of TFB-BD and Tp-BD membranes for the protic and aprotic solvents. UV-vis spectra of dye solutions of (b) Alcian blue (AB), (c) brilliant blue R (BBR), (d) methyl blue (MB), and (e) rhodamine B (RhB) before and after the filtration through TFB-BD-DMBS(I) and Tp-BD-DMBS(I) membranes, respectively. Digital photos of (f) AB, (g) BBR, (h) MB, and (i) RhB dye solutions before and after filtration, respectively.

tension would impede the monomer diffusion and the following spread-assembly process, resulting in the insufficient growth of the membrane with the thinnest thickness (Figure 5c), while for $[P_{4444}][Ma]$ with a relatively low viscosity and interface tension, a higher concentration of monomers would diffuse into the interface zone to trigger an intense bonding reaction and lead to the excessive growth of membrane with a loose stacking of irregular nanoparticles (Figure 5d). Particularly, the proper viscosity and interface properties of $[P_{4444}][DMBS]$ supplied a preferable condition for the diffusion-reaction equilibrium and further spread-assembly process to realize a thin, homogeneous COF film with good crystallinity (Figure 5e).

3.4. Permeation and Separation Performance Evaluation of COF Membrane. To evaluate the permeation and rejection performances of the synthesized COF membranes,

the solvent filtration experiments were first employed. The permeability of COF membranes to a series of protic (water, methanol, and ethanol) and aprotic (acetone and acetonitrile) solvents was investigated. As indicated in Figure 6a and Table S3, all COF membranes exhibited excellent permeability to protic solvents. For illustration, the permeance of TFB-BD-DMBS(I) was high up to $563.93 \text{ L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$ for water, $523.87 \text{ L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$ for methanol, and $469.60 \text{ L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$ for ethanol. Interestingly, it was observed that all of the COF membranes showed higher permeance toward aprotic solvents. The permeance values of TFB-BD-DMBS(I) for acetone and acetonitrile were 878.30 and $672.57 \text{ L}\cdot\text{m}^{-2}\cdot\text{h}^{-1}\cdot\text{bar}^{-1}$, respectively. The improved membrane permeability to aprotic solvents could be attributed to their lower viscosity as well as the weaker dipole-dipole interactions between aprotic solvents and the groups on the pores of COFs.⁵¹ Among these

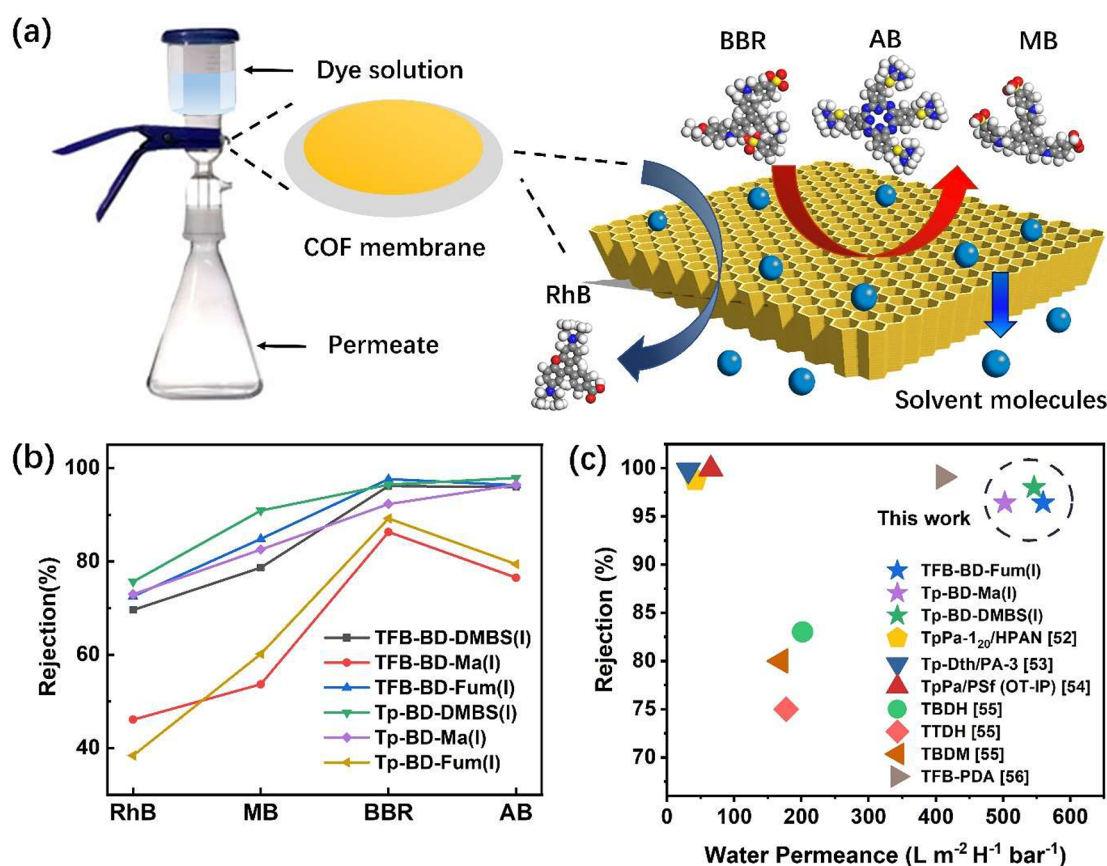


Figure 7. Rejection evaluation of TFB-BD and Tp-BD membranes. (a) Schematic representation of the molecule separation using COF membranes. (b) Rejection rate (%) of TFB-BD and Tp-BD COF to dyes of different molecular sizes in water. (c) Performance comparison of various nanofiltration membranes.

COF membranes, TFB-BD-DMBS(I) and Tp-BD-DMBS(I) which possessed better crystallinity exhibited a more outstanding permeation performance, demonstrating that the crystalline nanostructures provided hyperchannels for the rapid transportation of solvents.

Furthermore, the selective sieving performance of COF membranes for organic dyes with varying molecular sizes and charged properties including alcian blue (AB), brilliant blue R (BBR), methyl blue (MB), and rhodamine B (RhB) was evaluated. The chemical structures and detailed properties of the used organic dyes are listed in Table S4. As analyzed in Figure 5, the COF membranes synthesized at [P₄₄₄₄]-[DMBS]-based ATPSs interface showed a thin, homogeneous COF film with good crystallinity, thus were selected for further study. Figure 6b–e shows the UV–vis spectra of dye solutions before and after filtering. The membrane TFB-BD-DMBS(I) with near neutral nanochannels and Tp-BD-DMBS(I) with relative anionic nanochannels both exhibited high rejection rate for AB and BBR with large molecular size, illustrating that the dye rejection was dominated by size sieving when the dye molecule was larger than the channel size. Especially for Tp-BD-DMBS(I), the rejection rate for cationic dye AB was up to 98.0% despite the presence of electrostatic interactions between AB and the negative COF channels (Figure S14). Instead, the rejection rate of MB with negative charges but smaller molecular size decreased to 90.9% despite the synergistic effect of electrostatic repulsion, further suggesting that the rejection was governed by size sieving (Figure 6d). As a comparison, all COF membranes could only partially reject

RhB due to its smaller size than the pore size of the COFs (Figure 6e). The above results were further confirmed by comparison of the digital photographs of dye solutions before and after filtering (Figures 6f–i and S17). The schematic representation of the molecular separation using as-synthesized COF membranes is displayed in Figure 7a. Benefiting from the excellent crystallization and restricted pore structures, the TFB-BD and Tp-BD COF membranes prepared at [P₄₄₄₄]-[DMBS]-based ATPSs exhibited an excellent solvent permeability with remarkable rejection rate for dye molecules larger than the pore size of COF membranes (Figure 7b). The molecular weight cut off (MWCO) was determined to be 820 Da for TFB-BD-DMBS(I) and 800 Da for Tp-BD-DMBS(I) based on 90% dye rejection (Figure S18).

The recyclability of TFB-BD-DMBS(I) and Tp-BD-DMBS(I) membranes for permeance and BBR rejection was performed after rinsing by methanol. As displayed in Figure S19, the permeabilities of TFB-BD-DMBS(I) and Tp-BD-DMBS(I) membrane were more than 500 L·m⁻²·h⁻¹·bar⁻¹ over 5 cycles. Furthermore, the rejection rates of the recycled TFB-BD-DMBS(I) membrane for BBR was higher than 94.4% even after 5 cycles, and the rejection performance of the recycled Tp-BD-DMBS(I) membrane was negligibly changed, indicating their potential for long-term practical applications. To highlight the performance of the as-synthesized TFB-BD and Tp-BD COF membranes, water permeability and dye rejection performance for AB were compared with those of other membranes with similar interfacial synthesis in the previous works (Figure 7c and Table S5). It can be seen that

most of the separation membranes failed to combine high permeability and excellent dye rejection simultaneously.^{52–56} For the selected TFB-BD-Fum(I), Tp-BD-DMBS(I), and Tp-BD-Ma(I), the water permeance can be achieved at approximately 500 L·m⁻²·h⁻¹·bar⁻¹ with a comparable dye rejection. Such superior separation performance together with high permeability of our membranes generated from the unique well-ordered pore structure, strongly confirming that amide-linked COF membranes hold significant promise for high-performance membrane applications.

4. CONCLUSION

In summary, an IL-based aqueous two-phase interfacial polymerization strategy was developed to fabricate crystalline self-standing COF membranes. Through the simultaneous activation of aldehyde monomers by the bottom IL-rich phase and the amine monomers by a basic catalyst in the upper water-rich phase, the Schiff-base reaction at the IL-based aqueous biphasic interface was significantly enhanced. Compared with the traditional organic solvent–water interface, the IL-based aqueous biphasic was more stable, and its physicochemical properties can be tunable by the anionic structures of ILs. The diffusion control of monomers by high viscosity of ILs and the proper interface properties of biphasic interface resulted in the formation of thorough COF membranes with high crystallinity. Owing to the highly ordered pore structures, TFB-BD-DMBS(I) and Tp-BD-DMBS(I) exhibited excellent solvent permeability and reasonable rejection rate for dye molecules. As a breakthrough to fabricate a COF membrane at the IL-based aqueous biphasic interface, we expected that our work would provide a new direction for the green membrane manufacturing strategy.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.3c01939>.

Physicochemical properties and ¹H NMR spectra of the ILs; phase diagrams for IL-based ATPSs; the state of final COF products; synthesis of TFB-BD and Tp-BD COF; FT-IR spectra, PXRD patterns, pore size distribution, and top-view SEM images of TFB-BD(H) and Tp-BD(H); PXRD patterns of TFB-BD-single activation and Tp-BD-single activation; N₂ adsorption–desorption isotherms of TFB-BD and Tp-BD; TEM images of TFB-BD-DMBS(I) and Tp-BD-DMBS(I) films growth process; PXRD patterns of TFB-BD-DMBS(I) with different amine monomer activators; permeance and dye separation performance of the COF membranes; dye molecules and their properties; zeta potential, MWCO curve and cycle studies of TFB-BD-DMBS(I) and Tp-BD-DMBS(I) membranes (PDF)

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Author Contributions

L.L.: conceptualization, investigation, writing—original draft. Y.W.: characterization, discussion. W.R.: investigation, discussion. X.G.: investigation, characterization, discussion. L.Z.: supervision, funding acquisition. F.L.: conceptualization, supervision, funding acquisition, writing—review and editing.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

IL, ionic liquid; ATPSs, aqueous two-phase systems; COF, covalent organic framework; UCST, upper critical solution temperature; LCST, lower critical solution temperature

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